$$\varepsilon_i = -\frac{d\varphi}{dt}; \quad d\varphi = B\,dS, \quad dS = dy \cdot 2x$$

$$d\varphi = B\,dy \cdot 2x, \quad a = \frac{dv}{dt}$$

$$\varphi = \frac{2B}{\sqrt{k}} \int \sqrt{y}\,dy = \frac{2B}{\sqrt{k}} \cdot \frac{y^{\frac{3}{2}}}{\frac{3}{2}} + C^{=0}$$

$$\int dv = \int a\,dt;$$

$$v = at + C = 0$$

$$v = \frac{dy}{dt}; \quad dy = at\,dt$$

$$y = a \int t\,dt = \frac{at^2}{2}$$

$$\varphi(t) = \frac{2B}{\sqrt{k}} \cdot \frac{2(at^2)^{\frac{3}{2}}}{2^{\frac{3}{2}} \cdot 3} = \frac{\sqrt{2}\,B t^3 a^{\frac{3}{2}}}{\sqrt{k} \cdot 3}$$